



Predictive Modeling of Root-Zone Variables Using Greenhouse Data

Students: Eden Eliyahu, Oren Chaushu

Supervisors: Dr. Avner Priel, Dr. Alaa Jamal

01 Motivation

- Root-zone pH governs nutrient availability and EC reflects salinity and fertilizer load - both directly drive crop health and yield.
- Manual laboratory pH/EC samples are accurate but sparse, slow and reactive, leaving long blind gaps between measurements.
- Continuous probes drift, need frequent calibration and sample only a single point of the root zone.
- A machine-learning soft sensor fills these gaps, enabling proactive irrigation and fertigation decisions.

02 Project Goal

- Build a machine-learning soft sensor that continuously estimates root-zone pH and EC between sparse manual lab samples.
- Deliver reliable interpolated values so growers act on pH or salinity drift early - supporting timely irrigation and fertigation and keeping the root zone in its optimal band.

03 Data

16,682

10-min rows

109

pH / EC labels

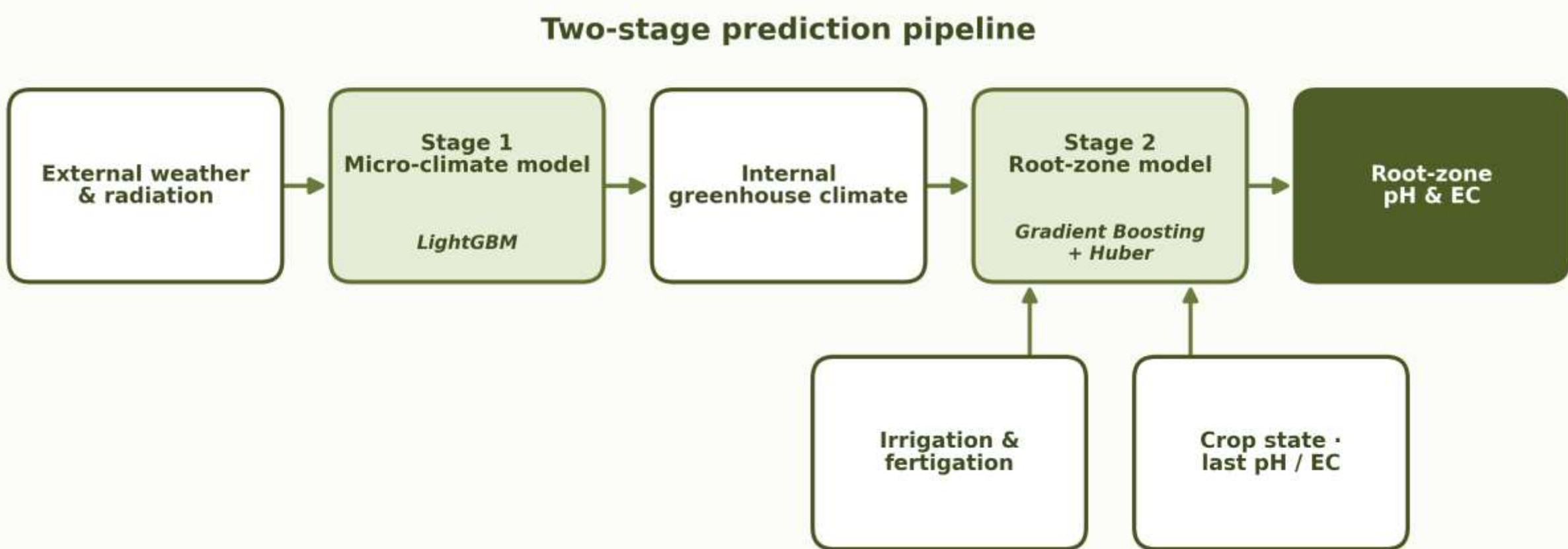
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input features

- Unified 10-minute master dataset · May 29 – Sep 21 2025.
- External weather & solar radiation (open-field station).
- Greenhouse climate - air temperature, RH, radiation, ET_o .
- Irrigation & fertigation - volumes, EC, acid dosing.
- Crop-state / canopy features - growth stage, canopy load.
- Manual laboratory pH & EC - sparse target labels.

04 Method

- Two-stage soft sensor - Stage 1 predicts greenhouse micro-climate from external weather; Stage 2 predicts root-zone pH & EC.
- Models the $t_0 \rightarrow t_1$ interval change rather than labelling every timestamp.



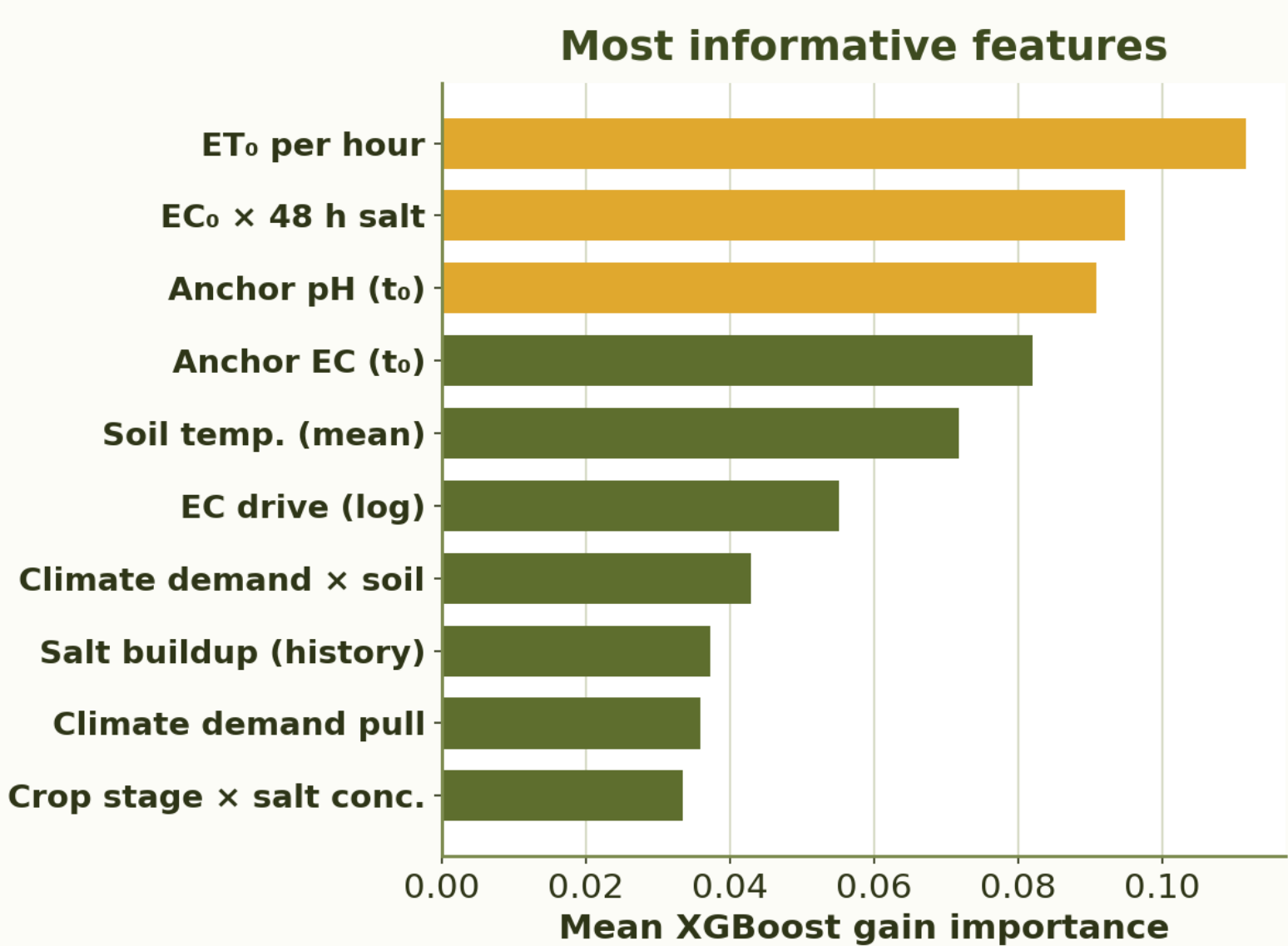
05 Algorithms & Model Choice

Gradient Boosting

Performs well on tabular greenhouse data; captures nonlinear interactions between climate, irrigation, fertigation, crop state and previous pH/EC. Suited to the hybrid interpolation / extrapolation task.

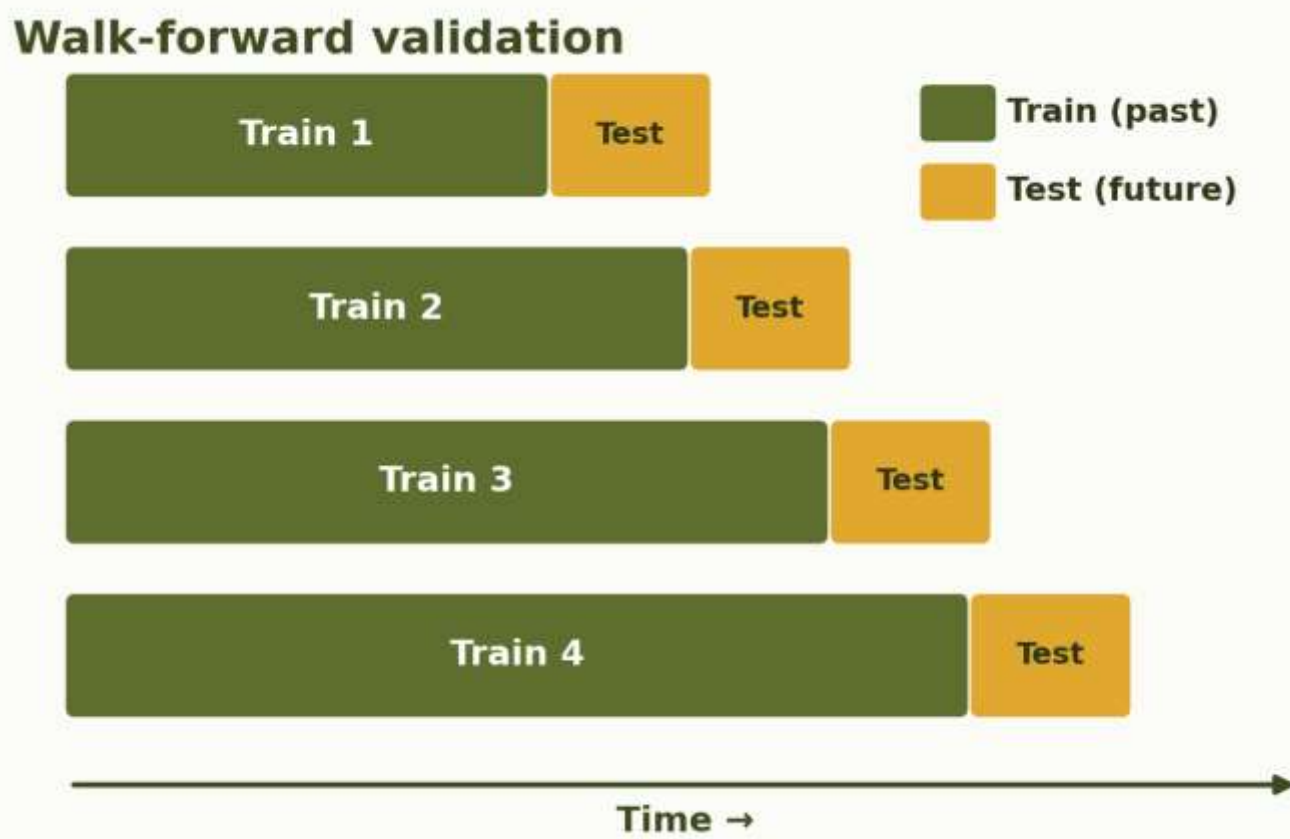
Huber Regressor

Robust to outliers and the sharp changes caused by fertigation, acid injection or salinity shifts. A stable, complementary model that handles noisy and extreme target behavior.



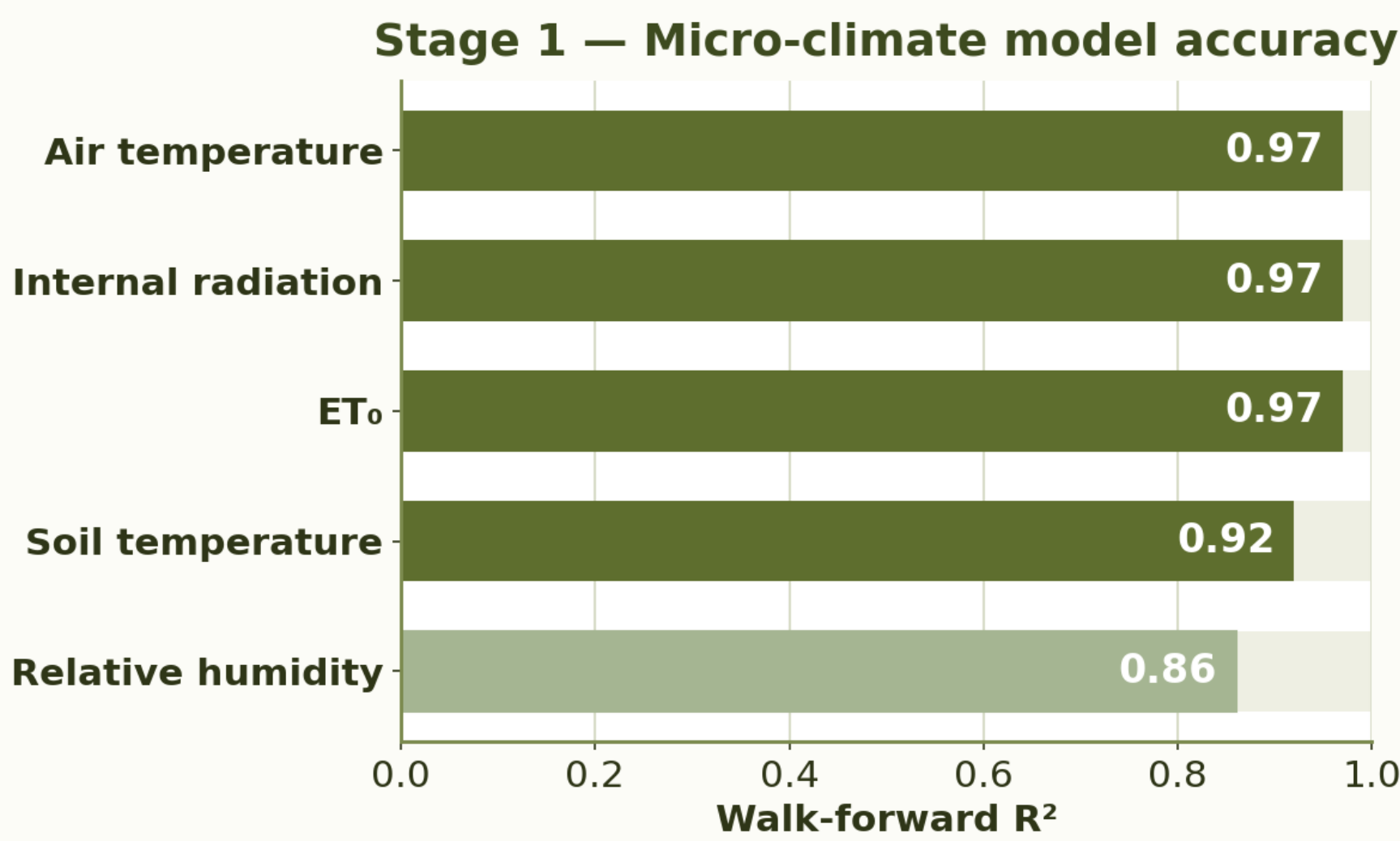
06 Validation Strategy

- Walk-forward validation: train only on past observations, predict future samples.
- Skipped-interval holdout: remove selected target intervals, predict them later.
- Metrics interpreted per target - pH (units) and EC (mS/cm).



07 Micro-Climate Model Results

- Estimates internal greenhouse climate from external weather & radiation.
- Feeds continuous climate features (ET_o , temperature, radiation, RH, soil temp) to the root-zone model.



08 Root-zone Model Results

0.33

pH MAE

0.93

pH R^2

+62%

pH MAE gain

0.127

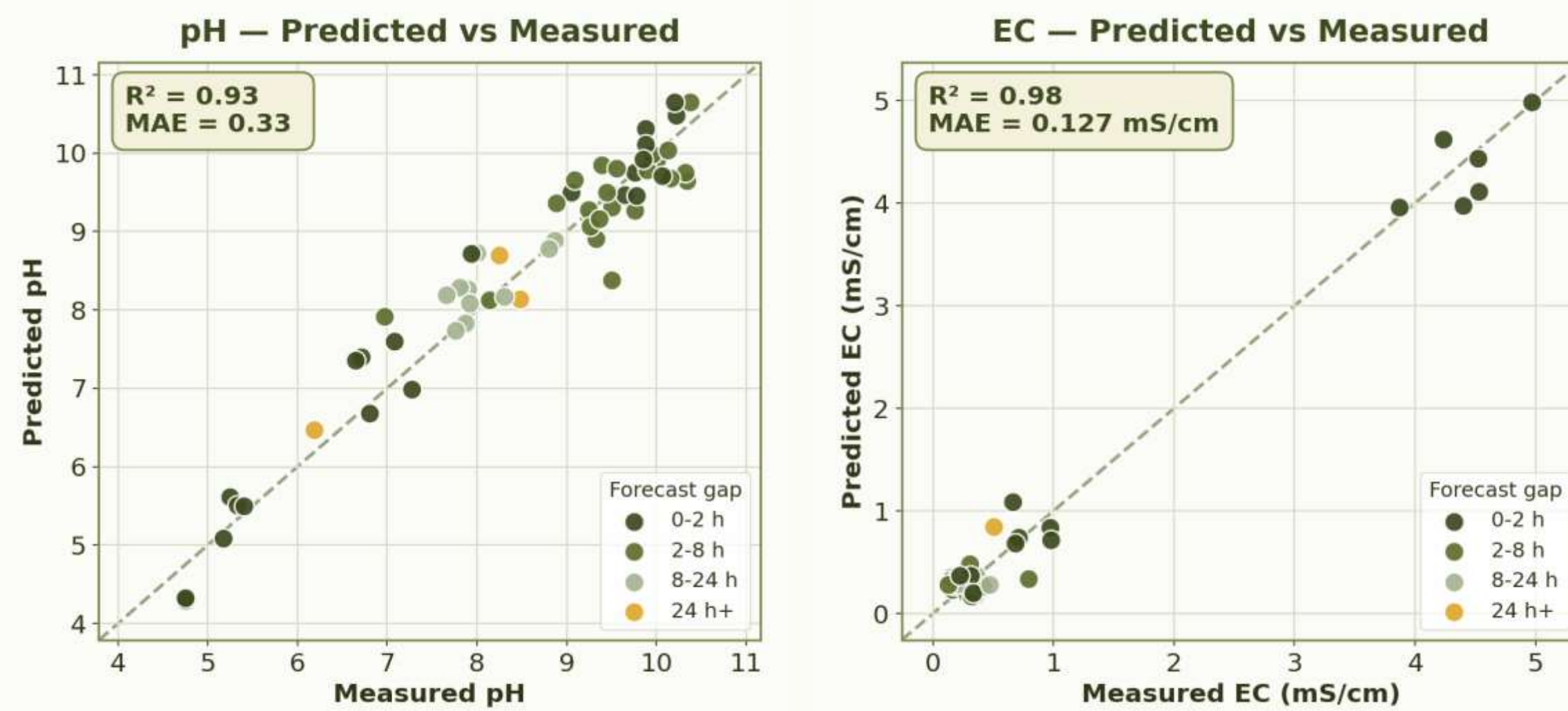
EC MAE

0.98

EC R^2

+35%

EC MAE gain



09 Conclusions

- Acts as a soft sensor, estimating root-zone pH & EC between sparse manual lab samples.
- The hybrid two-stage approach improves accuracy over the last-measurement baseline (+62% pH, +35% EC).
- Designed for decision support - guiding irrigation & fertigation - not autonomous control.

10 Challenges & Limitations

- Only 109 manual pH/EC labels against 16,682 dense sensor records.
- No continuous ground-truth pH/EC between lab samples.
- Irregular sampling intervals complicate interval modelling.
- Higher uncertainty in rare or extreme fertigation / salinity cases.
- Limited generalization — data from a single season and one crop.
- More seasons, crops & greenhouse conditions needed for stronger validation.